Bhavesh Verma

+91 7067419231 | bhaveshverma7890@gmail.com
B.Tech. (Hons.) in Mining Engineering and Master of Petroleum Engineering, IIT Kharagpur | Micro-Specialization in AI & ML

Educational Qualifications

Year	Degree	Institute/School	CGPA/Percentage
2023–2028	M.Tech. Dual Degree (5Y)	IIT Kharagpur	8.30/10
2022	Higher Secondary Examination (CBSE)	Shri Shankracharya Vidyalaya	86%
2020	Secondary Examination (CGBSE)	Sun Shine Hr. Sec. School	91%

Experience

Data Science Intern March 2026 – May 2026
Red Panda Games *Bengaluru, India*

- Architected a Generative AI pipeline integrating **Flux**, **SD1.5**, and **Trellis-4B** to synthesize game-ready 3D meshes from prompts, orchestrating **ControlNet** with dynamic scaling and letterbox-padded **OpenPose** skeletons for accurate body proportions
- Optimized **AWS** GPU spot instances (L4/L40S) across 40GB+ workloads, resolving critical OOM crashes via CPU memory offloading and Linux Swap
- Engineered Python microservices with **Celery** and **Redis** to decouple 3D rendering into dedicated workers, and streamlined mesh assets for Auto-Rig Pro using **U2-Net** background removal, quad-dominant retopology, and 1M-triangle decimation

Machine Learning Intern May 2025 – July 2025
Prof. Kuntal Roy, IISER Bhopal *Bhopal, India*

- Conducted research in **NLP** and Generative AI, formulating scalable end-to-end pipelines using **Logistic Regression**, **Naive Bayes**, and word embedding methodologies
- Built deep learning models leveraging **LSTMs**, **GRUs**, and Transformer architectures such as **BERT** and **T5**; developed NLP applications including sentiment analysis, POS tagging, and machine translation using fine-tuned pretrained Transformer models

Projects

ControlPlane.ai: AI Governance Middleware | *LangGraph, LiteLLM, FAISS, XGBoost, Neo4j* May 2026 – June 2026

- Architected a 9-node **LangGraph** state machine that guards, routes, answers, and dual-audits every LLM response in parallel, hitting a sub-**10s** p95 latency budget (verified via automated benchmark) and **5–8s** end-to-end on warm infrastructure
- Built an agentic router across **5** knowledge bases (keyword fast-path, LLM tool-calling, embedding fallback) feeding hybrid retrieval that fuses **FAISS** dense and **BM25** sparse search per KB via **Reciprocal Rank Fusion** (k=60)
- Engineered parallel audit branches – a 12-feature **XGBoost** hallucination classifier with **RAGAS** judge and a 3-model toxicity ensemble (Detoxify, ToxicBERT, RoBERTa) backed by a **Neo4j** compliance retriever, capping latency at $<$7.4s; served via a **LiteLLM** gateway with failover and semantic caching, validated by **28** tests and **LangSmith** tracing in a **Streamlit** dashboard

Building a Network Security System | *MongoDB Atlas, MLflow, DagsHub, Docker, AWS, CI/CD* Nov 2025 – Dec 2025

- Programmed a Network Security ML platform predicting phishing or safe URLs, enabling real-time threat classification via user-uploaded data
- Automated ETL pipeline using **MongoDB Atlas**, streamlined experiment tracking via **MLflow/DagsHub**, and reduced deployment latency by **40%** using Docker and CI/CD pipelines on AWS

E-Commerce Price Prediction | *DeBERTa-v3, Regex, NLP Feature Engineering* Jan 2026 – Feb 2026

- Devised a text-based regression model to predict product prices from catalog descriptions using advanced NLP feature engineering
- Leveraged a hybrid pipeline combining Regex feature extraction and **DeBERTa-v3** embeddings, achieving **42.22% SMAPE**; configured training with Huber Loss and dynamic padding, reducing inference latency by **30%**

Competitions & Achievements

Character Backstory Verification System | *National Finalist, Kharagpur Data Science Hackathon 2026* Dec 2025 – Jan 2026

- Designed a production-level **Retrieval-Augmented Generation (RAG)** pipeline for robust character backstory verification, integrating a hybrid retrieval system using BM25, FAISS dense embeddings, reciprocal rank fusion, and cross-encoder reranking
- Applied **DeBERTa-v3 NLI** to detect semantic text entailment/contradiction with confidence scores, achieving **81.25% accuracy** and **0.854 F1-score**

Predictive Profitability Framework | *Top Performer, American Express Campus Challenge 2026* July 2026 – Aug 2026

- Architected a hybrid ensemble blending a financial formula with **LightGBM/XGBoost** to predict profitability, achieving verified **91.33% accuracy**
- Derived a broad P&L profitability framework from 10K data research, incorporating 24% APR, 3% interchange, 96% URR economics
- Generated **2M+** Out-of-Fold predictions via 5-Fold CV, applying rank-normalization and SHAP to validate feature-driven, credit risk-based decisions

Technical Skills

Languages: Python (proficient), C, C++, CSS, HTML
Frameworks & Libraries: NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Streamlit, Flask, BentoML, FastAPI
GenAI & ML: LangGraph, LangChain, LiteLLM, Hugging Face, Flux, Trellis, Stable Diffusion, RAG, ChromaDB, FAISS, XGBoost, RAGAS, Langfuse, LLM-as-Judge
Tools: Git, GitHub, Docker, MATLAB, AWS, Neo4j, DagsHub, MLflow, LangSmith, Excel, VS Code, Canva, ETL Pipeline, Figma

Additional Information

Certifications: Machine Learning, Deep Learning & NLP Specializations; Data Science, ML, DL & NLP Bootcamp (Krish Naik)
Extracurriculars: Inter-Hall Championship – Product Design; 3-time Silver Medalist, Inter-Hall Yoga Championship